

TRIGGER IN SQL

1) AFTER INSERT Trigger :

```
mysql> CREATE TABLE Student (  
->     student_id INT AUTO_INCREMENT PRIMARY KEY,  
->     student_name VARCHAR(100) NOT NULL,  
->     major VARCHAR(50)  
-> );  
ERROR 1046 (3D000): No database selected  
mysql> CREATE DATABASE CollegeDB;  
Query OK, 1 row affected (0.01 sec)  
  
mysql> USE CollegeDB;  
Database changed  
mysql> CREATE TABLE Student (  
->     student_id INT AUTO_INCREMENT PRIMARY KEY,  
->     student_name VARCHAR(100) NOT NULL,  
->     major VARCHAR(50)  
-> );  
Query OK, 0 rows affected (0.03 sec)  
  
mysql> CREATE TABLE Student_Count (  
->     total_students INT DEFAULT 0  
-> );  
Query OK, 0 rows affected (0.02 sec)  
  
mysql> INSERT INTO Student_Count (total_students) VALUES (0);  
Query OK, 1 row affected (0.02 sec)  
  
mysql> DELIMITER //  
mysql>  
mysql> CREATE TRIGGER after_student_insert  
-> AFTER INSERT ON Student  
-> FOR EACH ROW  
-> BEGIN  
->     UPDATE Student_Count  
->     SET total_students = total_students + 1;  
-> END //  
Query OK, 0 rows affected (0.01 sec)  
  
mysql>  
mysql> DELIMITER ;  
mysql> INSERT INTO Student (student_name, major) VALUES ('Alice Smith', 'Computer Science'  
-> );
```

```

mysql> INSERT INTO Student (student_name, major) VALUES ('Alice Smith', 'Computer Science'
-> );
Query OK, 1 row affected (0.01 sec)

mysql> SELECT * FROM Student_Count;
+-----+
| total_students |
+-----+
|                1 |
+-----+
1 row in set (0.00 sec)

mysql> INSERT INTO Student (student_name, major) VALUES ('mohith', 'Computer Science')
-> INSERT INTO Student (student_name, major) VALUES ('many', 'general Science');
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to y
(student_name, major) VALUES ('many', 'general Science')' at line 2
mysql> INSERT INTO Student (student_name, major) VALUES ('mohith', 'Computer Science'),
-> INSERT INTO Student (student_name, major) VALUES ('many', 'general Science');
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to y
(student_name, major) VALUES ('many', 'general Science')' at line 2
mysql> INSERT INTO Student (student_name, major) VALUES ('mohith', 'Computer Science');
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO Student (student_name, major) VALUES ('many', 'general Science');
Query OK, 1 row affected (0.00 sec)

mysql> SELECT * FROM Student_Count;
+-----+
| total_students |
+-----+
|                3 |
+-----+
1 row in set (0.00 sec)

```

2) AFTER UPDATE Trigger :

```
mysql> USE CompanyDB;
Database changed
mysql> CREATE TABLE Employee (
->     emp_id INT AUTO_INCREMENT PRIMARY KEY,
->     emp_name VARCHAR(100) NOT NULL,
->     salary DECIMAL(10, 2) NOT NULL);
Query OK, 0 rows affected (0.03 sec)

mysql> CREATE TABLE Salary_Audit (
->     audit_id INT AUTO_INCREMENT PRIMARY KEY,
->     emp_id INT,
->     old_salary DECIMAL(10, 2),
->     new_salary DECIMAL(10, 2),
->     change_time TIMESTAMP DEFAULT CURRENT_TIMESTAMP
-> );
Query OK, 0 rows affected (0.03 sec)

mysql> DELIMITER //
mysql>
mysql> CREATE TRIGGER after_salary_update
-> AFTER UPDATE ON Employee
-> FOR EACH ROW
-> BEGIN
-> ^C
mysql> CREATE TRIGGER after_salary_update
-> AFTER UPDATE ON Employee
-> FOR EACH ROW
-> BEGIN
-> IF OLD.salary <> NEW.salary THEN
->     INSERT INTO Salary_Audit (emp_id, old_salary, new_salary)
->     VALUES (OLD.emp_id, OLD.salary, NEW.salary);
-> END IF;
-> END //
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
mysql> INSERT INTO Employee (emp_name, salary) VALUES ('John Doe', 50000.00);
```

```

mysql> INSERT INTO Employee (emp_name, salary) VALUES ('Mohith', 65000.00);
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO Employee (emp_name, salary) VALUES ('Manya', 58000.00);
Query OK, 1 row affected (0.00 sec)

mysql> INSERT INTO Employee (emp_name, salary) VALUES ('Alice Smith', 45000.00);
Query OK, 1 row affected (0.00 sec)

mysql> INSERT INTO Employee (emp_name, salary) VALUES ('Bob Jones', 72000.00);
Query OK, 1 row affected (0.01 sec)

mysql> UPDATE Employee SET salary = 75000.00 WHERE emp_name = 'Mohith';
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> UPDATE Employee SET salary = 62000.00 WHERE emp_name = 'Manya';
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> UPDATE Employee SET salary = 74500.00 WHERE emp_name = 'Shobi';
Query OK, 0 rows affected (0.00 sec)
Rows matched: 0  Changed: 0  Warnings: 0

mysql> SELECT * FROM Salary_Audit;
+-----+-----+-----+-----+-----+
| audit_id | emp_id | old_salary | new_salary | change_time |
+-----+-----+-----+-----+-----+
| 1 | 1 | 50000.00 | 55000.00 | 2026-09-29 12:02:36 |
| 2 | 2 | 65000.00 | 75000.00 | 2026-09-29 12:03:56 |
| 3 | 3 | 58000.00 | 62000.00 | 2026-09-29 12:04:09 |
+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)

```

3) AFTER DELETE Trigger :

```
mysql> USE HRDB;
Database changed
mysql> CREATE TABLE Employee (
  ->     emp_id INT AUTO_INCREMENT PRIMARY KEY,
  ->     emp_name VARCHAR(100) NOT NULL,
  ->     department VARCHAR(50) NOT NULL,
  ->     salary DECIMAL(10, 2) NOT NULL
  -> );
Query OK, 0 rows affected (0.03 sec)

mysql> CREATE TABLE Deleted_Employee_Log (
  ->     log_id INT AUTO_INCREMENT PRIMARY KEY,
  ->     emp_id INT,
  ->     emp_name VARCHAR(100),
  ->     department VARCHAR(50),
  ->     deleted_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
  -> );
Query OK, 0 rows affected (0.03 sec)

mysql> DELIMITER //
mysql>
mysql> CREATE TRIGGER prevent_management_deletion
  -> BEFORE DELETE ON Employee
  -> FOR EACH ROW
  -> BEGIN
  ->     -- Check if the employee belongs to Management
  ->     IF OLD.department = 'Management' THEN
  ->         SIGNAL SQLSTATE '45000'
  ->         SET MESSAGE_TEXT = 'Error: Employees in the Management department cannot be deleted!';
  ->     END IF;
  -> END //
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
mysql> DELIMITER //
mysql>
mysql> CREATE TRIGGER log_deleted_employee
  -> AFTER DELETE ON Employee
  -> FOR EACH ROW
  -> BEGIN
```

```

-> INSERT INTO Deleted_Employee_Log (emp_id, emp_name, department)
-> VALUES (OLD.emp_id, OLD.emp_name, OLD.department);
-> END //
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
mysql> INSERT INTO Employee (emp_name, department, salary) VALUES ('Mohith', 'Management', 95000.00);
Query OK, 1 row affected (0.00 sec)

mysql> INSERT INTO Employee (emp_name, department, salary) VALUES ('Alice Smith', 'IT', 60000.00);
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO Employee (emp_name, department, salary) VALUES ('Bob Jones', 'HR', 55000.00);
Query OK, 1 row affected (0.00 sec)

mysql> DELETE FROM Employee WHERE emp_name = 'Mohith';
ERROR 1644 (45000): Error: Employees in the Management department cannot be deleted!
mysql> DELETE FROM Employee WHERE emp_name = 'Alice Smith';
Query OK, 1 row affected (0.01 sec)

mysql> SELECT * FROM Employee;
+-----+-----+-----+-----+
| emp_id | emp_name | department | salary |
+-----+-----+-----+-----+
|      1 | Mohith   | Management | 95000.00 |
|      3 | Bob Jones | HR         | 55000.00 |
+-----+-----+-----+-----+
2 rows in set (0.00 sec)

mysql> SELECT * FROM Deleted_Employee_Log;
+-----+-----+-----+-----+-----+
| log_id | emp_id | emp_name | department | deleted_at |
+-----+-----+-----+-----+-----+
|      1 |      2 | Alice Smith | IT         | 2026-09-29 12:08:12 |
+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)

```